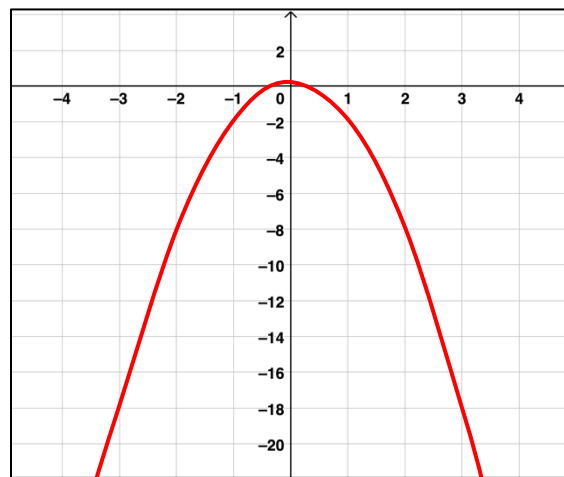


Use the function $y = -2x^2$ to answer problems 1–4.

1. Complete the table of values to represent the function.

x	y	
-3	-18	$-2(-3)^2 = -18$
-1	-2	
0	0	$0 = -2x^2 \quad x = 0$
1	-2	$-2(1)^2 = -2$
± 2	-8	$-8 = 2x^2 \quad x = \pm 2$

2. Graph the points from the table and create a rough sketch of the function by connecting the points.



3. In the function $y = -2x^2$, which x -values will produce the y -value of -50 ?

$$\begin{aligned} \frac{-50}{-2} &= \frac{-2x^2}{-2} \\ 25 &= x^2 \\ x &= \pm 5 \end{aligned}$$

4. Did the function $y = -2x^2$ represent a linear function? Using your conjecture from the lesson why do you think this is true?
No because x^2 has positive and negative x -values that results in the same y -value.

Create a table of values for each function. **Answers may vary.**

5. $y = x^2 + 2$

x	y
-3	11
-1	3
0	2
1	3
3	11

$(-3)^2 + 2$
 $(-1)^2 + 2$
 $(0)^2 + 2$
 $(1)^2 + 2$
 $(3)^2 + 2$

6. $y = 3x - 10$

x	y
-4	-22
-2	-16
0	-10
2	-4
4	2

$3(-4) - 10$
 $3(-2) - 10$
 $3(0) - 10$
 $3(2) - 10$
 $3(4) - 10$

7. $y = x^3 - 3$

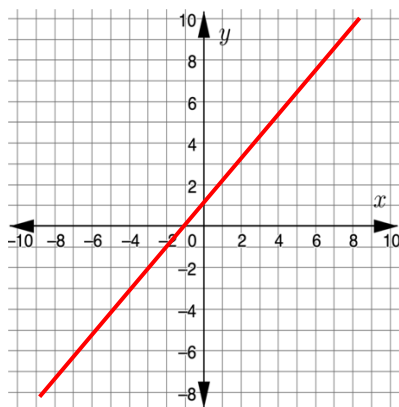
x	y
-1	-4
0	-3
1	-2
2	5
3	24

$(0)^3 - 3$
 $(-1)^3 - 3$
 $(1)^3 - 3$
 $(2)^3 - 3$
 $(3)^3 - 3$

Create a graph for each function.

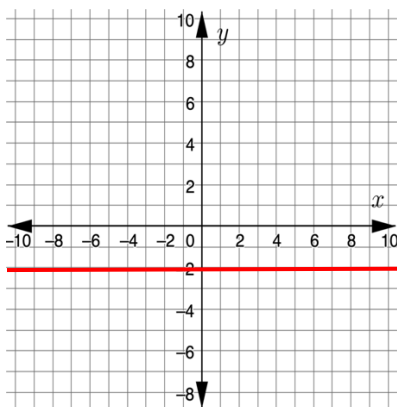
8.

x	y
-2	-1
-1	0
0	1
1	2
2	3



9.

x	y
-2	-2
-1	-2
0	-2
1	-2
2	-2



10.

x	y
-2	10
-1	7
0	4
1	1
2	-2

